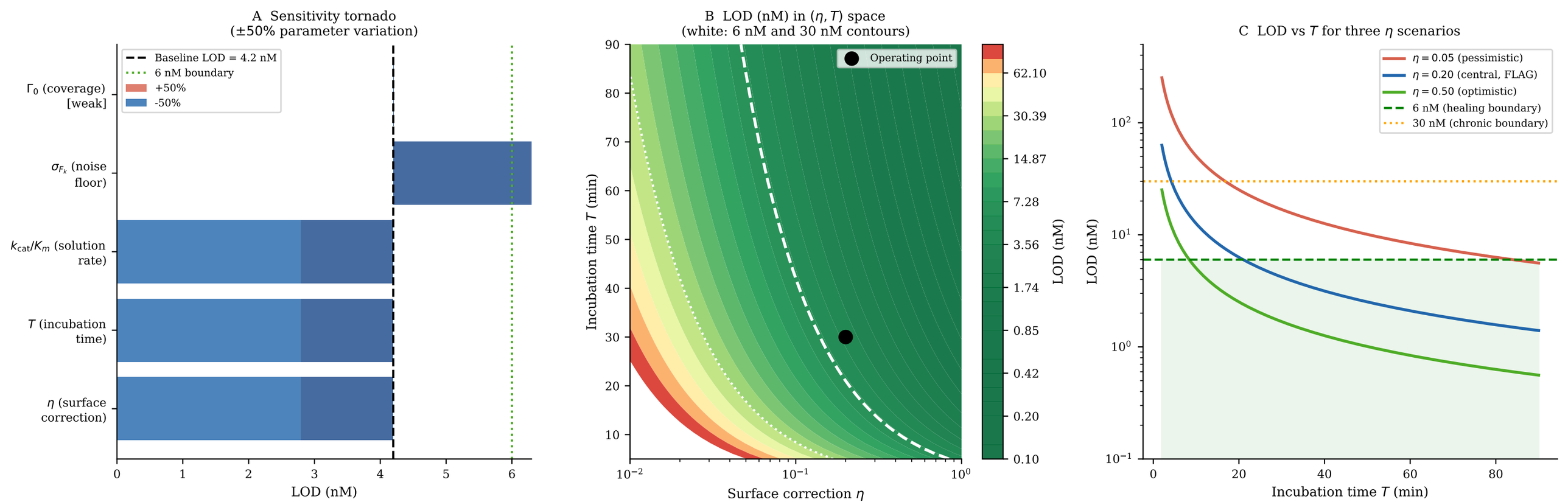

A Sensitivity tornado
(±50% parameter variation)
Baseline LOD = 4.2 nM
6 nM boundary
+50%
-50%
Γ0 (coverage) [weak]
σFk (noise floor)
kcat/Km (solution rate)
T (incubation time)
η (surface correction)
0
1
2
3
4
5
6
LOD (nM)
B LOD (nM) in (η, T) space
(white: 6 nM and 30 nM contours)
Operating point
Incubation time T (min)
Surface correction η
10−2
10−1
100
62.10
30.39
14.87
7.28
3.56
1.74
0.85
0.42
0.20
0.10
LOD (nM)
C LOD vs T for three η scenarios
η = 0.05 (pessimistic)
η = 0.20 (central, FLAG)
η = 0.50 (optimistic)
6 nM (healing boundary)
30 nM (chronic boundary)
LOD (nM)
Incubation time T (min)